PERSONAL INFORMATION

Md Abu Sayed

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WORK EXPERIENCE

Jun 2019–Present

Lecturer

The Millennium University, Dhaka (Bangladesh)

Department of Computer Science and Engineering

- **Instructed undergraduate courses**: Structured Programming, Data Structure, Artificial Intelligence & Neural Networks, Object-Oriented Analysis & Design, Software Engineering, etc.
- **Responsibilities**: Lecture planning, taught and instructed courses, assessing students, holding office hours, invigilating examinations, supervising application development projects, etc.
- **Course Coordinator** of the Department since December 2019
- Spokeperson of the Sports Teams in "The Bangabandhu Sports Tournament 2020"

EDUCATION AND TRAINING

Dec 2014–Jan 2019

Bachelor of Science in Computer Science & Engineering

Khulna University, Khulna (Bangladesh)

CGPA: 3.31 out of 4

Programming Courses

- Structured Programming, Data Structure, Algorithms, Object-Oriented Programming, etc.

Other Major Courses

- Database, Software Engineering, Computer Networks
- Microprocessor & Micro-controller, Image Processing, Computer Graphics
- Artificial Intelligence, Data Mining , Pattern Recognition, Applied Probability & Queuing Theory

RESEARCH AND PROJECTS

Research Interests

Computer Vision, Machine Learning in Autonomous Driving, Medical Image Analysis, Health-care etc.

Publications

1. **Sayed, M.A.**, Saha, S., Rahaman, G.A., Ghosh, T.K. and Kanagasingam, Y., An innovate approach for retinal blood vessel segmentation using mixture of supervised and unsupervised methods. IET Image Processing. 2020;1-11. [**PDF**]

2. **Sayed M.A.**, Saha S., Rahaman G.M.A., Ghosh T.K., Kanagasingam Y. (2019) **A Semi-supervised Approach to Segment Retinal Blood Vessels in Color Fundus Photographs**. In: Riaño D., Wilk S., ten Teije A. (eds) Artificial Intelligence in Medicine. AIME 2019. Lecture Notes in Computer Science, vol 11526. Springer, Cham. [**PDF**]

3. Ghosh T.K., Saha S., Rahaman G.M.A., **Sayed M.A.**, Kanagasingam Y. (2019) **Retinal Blood Vessel Segmentation: A Semi-supervised Approach.** In: Morales A., Fierrez J., Sánchez J., Ribeiro B. (eds) Pattern Recognition and Image Analysis. IbPRIA 2019. Lecture Notes in Computer Science, vol 11868. Springer, Cham. [**PDF**]

Bachelor Thesis

Automated method to segment retinal blood vessels from color fundus photographs. [PDF]

Supervisor: Professor Dr. G M Atiqur Rahaman, **Co-supervisor**: Dr. Sajib Saha (CSIRO, Australia)

In this thesis, we propose a method to segment retinal blood vessel in presence of pathological lesion. The proposed method is also able to overcome the most of the other challenges such as segmentation in presence of central vessel reflex, crossover and bifurcation regions etc. To extract trainable feature, we propose a descriptor and compare the descriptor with a novel descriptor (SURF). We get nearly identical performance for the proposed descriptor respect to SURF descriptor.

Projects

1. Deep Learning Projects [Few included]

Python, Tensorflow, Keras

- Implementing CNN from Scratch with Python
- Multiple diseases detection from Chest X-ray using single DenseNet
- Brain Tumour auto-segmentation from 3D MRI images using U-Net
- Car Detection with Yolo Algorihtm
- Image Captioning with CNN and Bi-directional LSTM

2. EmotionAI: Facial Key Point Detection

Keras, Python | Project Link | Github

- A model based Convolutional Neural Network with residual blocks to predict facial key points

3. Local Haar Pattern (LHP): A Feature Descriptor for Biomedical Images

MATLAB | Project Link

4. Pathik: A Bi-pedal Walking Robot

Servo Motor, Servo Controller (Micro-controller) | Raw Walking Demo *(Later improved)*

PERSONAL SKILLS

Technical skills

- **AI/ Machine Learning and Computer Vision**: MATLAB, Python, Keras (Proficient), Tensorflow, OpenCV (Familiar)
- **Problem Solving**: C/C++ (Proficient), Java, C# (Familiar)
- **Web Development**: PHP, MySQL (Proficient), JavaScript, SQL (Familiar)
- ***Tools***: LaTeX, Microsoft Word/ PowerPoint, **Version Control**: Git

Organizational skills

- Organized and managed Gaming Section of **Inter-University CSE FEST** on behalf of CLUSTER; a computer club that arranges seminars, programming contests and publishes magazines, etc.
- Co-founder and volunteer of Muktomoncho, a community organization which addresses the mental health of nationwide students

Communication skills

- Developed excellent communication skills since
  - I have been conducting lectures and supervising Bachelor students for more than a year
  - I have worked with hundreds of university students as a part of **Muktomoncho**
- Improved collaboration skills through
  - Working as a Course Coordinator of my department for 10 months
  - working as a leader or member of several teams in different academic projects

ADDITIONAL INFORMATION

Standardized Test Scores

1. **IELTS**: **Band Score 7.5** (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 7.5)
2. **GRE**: **Overall 316** (Quant Reasoning: 160, Verbal Reasoning: 156, Analytical Writing: 3.0)

MOOCS & Trainings

1. Deep Learning Specialization (5 Courses) by Andrew NG (deeplearining.ai)

- **Courses**: Neural Network and Deep Learning, Hyperparameter Optimization, Structuring Machine Learning Projects, Convolutional Neural Networks, Sequence Models

2. Mathematics for Machine Learning Specialization

- Imperial College, London; Coursera, **Courses:** Linear Algebra

3. AI for Medicine by deeplearning.ai

- **Courses:** AI for Medical Diagnosis

4. Machine Learning A-Z™: Hands-On Python & R In Data Science

- 41.5 hrs., Udemy, Certificate Link

5. Industrial IoT Markets and Security, University of Colorado Boulder; Coursera

6. Japanese Language Training (April 2020 – December 2020, JLPT N4 Level)

References

References available on request.